Juan Pablo García

AI Developer · RAG, agents and LLM evaluation

Available now for full-time remote work

San Juan, Argentina (UTC−3) · jpgarcia2154@gmail.com · garciajp.com
linkedin.com/in/juan-pablo-garcia-guerrero · github.com/jpgg2105 · x.com/garciajpgg21

EXPERIENCE

AI Developer, Nodos Hub · Remote, part-time contract — Feb 2025 – Present

- Built a React and FastAPI analytics dashboard, now in production, tracking student attendance, performance and risk indicators across hundreds of schools for an education-sector client, with Excel export.
- Own the full deployment pipeline from GitLab to a cloud Linux server (Gunicorn, systemd, nginx) across separate dev and prod environments, with SSO against the client's identity system.
- Built LLM chatbots (Gemini, Pinecone) in PHP and Python, plus an automated lead-scraping and WhatsApp outreach pipeline using n8n, Airtable and the OpenAI API.

AI Developer, Glow Talent · Remote, part-time contract — Sep 2025 – Jul 2026

- Built a multi-tenant SaaS platform from scratch as the sole developer, from database design to deployment, using Python, Flask, PostgreSQL, SQLAlchemy and Alembic.
- Built retrieval-augmented generation (ChromaDB, Gemini) and LLM-based evaluation features.
- Implemented role-based access control with JWT authentication, and automated QA with headless-browser testing.

PROJECTS

Research agent and benchmark · research-agent · agent-benchmark · ai-document-analyst — Python, FastAPI, React, Docker

- A planner, researcher, critic and synthesizer loop on my RAG service, benchmarked against a single LLM call on 100 HotpotQA multi-hop questions: 86% vs 74% correct (+12 points, 95% CI +4 to +20), at about 6× the calls.

Retrieval ablation · retrieval-ablation — sentence-transformers, BM25, BEIR

- Measured each retrieval stage of the RAG service on 1,008 labeled questions: a one-line BM25 tokenizer fix raised recall@5 from 55% to 78%; the reranker took over 95% of query time, and 10 candidates did as well as 20. Fixes merged.

Prompt injection lab · agent-injection-lab — Gemini, tool-using agent

- Planted 36 injections in a tool-using assistant's documents and tested four defenses over 360 runs: a system-prompt warning cut successful attacks from 42% to 6%; an approval gate stopped every hijacking email, not leaks in the answer.

LoRA vs prompting · lora-vs-prompting — PEFT, Transformers, Colab T4

- Fine-tuned Qwen2.5-1.5B with LoRA against few-shot Gemini Flash-Lite on structured extraction at 100, 500 and 2,000 labeled examples: prompting won by over 20 points at 100; at 2,000 LoRA was within noise (64% vs 66% exact) with 29× fewer input tokens.

EDUCATION

Universidad de Palermo — Oct 2024 – Jul 2029 (expected)
B.Eng. in Artificial Intelligence Engineering; intermediate AI Analyst degree expected Dec 2026

University of Pennsylvania — Aug 2023 – May 2024
Full-year exchange, BSE Computer Science. GPA 3.88/4.0, 80% scholarship, completed with academic honors

Escuela Industrial D. F. Sarmiento — 2016 – 2022
Electronics Technician. GPA 9.40, graduated Valedictorian (Abanderado)

TECHNICAL SKILLS

- **AI / ML:** RAG, LLM integration (Gemini, OpenAI, Anthropic), vector search (ChromaDB, Pinecone, Qdrant), multi-agent systems, LLM evaluation, LoRA fine-tuning (PEFT, Transformers), PyTorch, scikit-learn
- **Backend and frontend:** Python, FastAPI, Flask, REST APIs, SQLAlchemy, Alembic, JWT, RBAC, OAuth; React, TypeScript, Vite, Tailwind CSS
- **Infrastructure:** PostgreSQL, Docker, GitHub Actions, Git, Linux, Gunicorn, systemd, nginx, Azure, pytest, Playwright
- **Languages:** Python, TypeScript, JavaScript, SQL, Java, PHP, C

HONORS AND LANGUAGES

- Bronze Medal, International Junior Science Olympiad (Qatar, 2019); Gold Medal, National Junior Science Olympiad (2019); 80% UPenn scholarship; 100% ITBA scholarship offer. Spanish (native), English (C2), French (A2), Italian (basic).